NESS SPACE

Mars Expedition — User Guide

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How to explore

NESS SPACE is an augmented-reality expedition across the Martian surface. A glowing pink fluorescent guide line connects 50 scannable waypoints — from the volcanoes of Tharsis, through the vast Valles Marineris canyon system, and across to the frozen northern plains of Utopia Planitia.

Follow the pink line from beacon to beacon. When a beacon comes into range, initiate a scan to log its coordinates to Mars Global Localization. Each scan reveals a description of the location and advances you toward the next point. You may also jump directly to any of the 50 sites from the Sites menu inside the expedition, or from this guide.

An expert AI guide named Ares can be unlocked for richer, real-time narration during your journey — 30 minutes of guidance cost 1.00 USD, payable through MetaMask.

Below is every scannable point on the expedition, in order, with its true Martian coordinates, elevation, and a description.

50 scannable points (in expedition order)

1 Olympus Mons

Lat 18.65° N · Lon 133.83° W · Elevation 21,229 m

The tallest volcano and mountain in the solar system — a 600 km-wide shield rising 21 km above the plains.

2 Ascraeus Mons

Lat 11.8° N · Lon 104.4° W · Elevation 18,225 m

The tallest of the Tharsis volcanoes, crowned by a complex summit caldera.

3 Pavonis Mons

Lat 0.8° N · Lon 112.5° W · Elevation 14,057 m

The 'Peacock Mountain', sitting on the equator with a marked summit pit.

4 Arsia Mons

Lat 8.5° S · Lon 120.1° W · Elevation 16,000 m

Southernmost Tharsis shield; its caldera is often shrouded in morning water-ice clouds.

5 Alba Mons

Lat 39.7° N · Lon 110.0° W · Elevation 6,800 m

A vast, low-sloped shield volcano unique to the northern hemisphere.

6 Uranius Mons

Lat 26.0° N · Lon 98.0° W · Elevation 3,500 m

A smaller Tharsis volcano flanked by fissures and lava channels.

7 Ceraunius Tholus

Lat 24.0° N · Lon 97.5° W · Elevation 5,500 m

A steep-sided volcanic cone cut by a deep, winding canyon.

8 Tharsis Tholus

Lat 13.5° N · Lon 90.0° W · Elevation 3,000 m

A fractured, collapsed-flank volcano east of the Tharsis rise.

9 Noctis Labyrinthus

Lat 9.0° S · Lon 102.0° W · Elevation "2,000 m

A maze of intersecting graben valleys marking the start of the canyon system.

10 Valles Marineris

Lat 13.9° S · Lon 59.2° W · Elevation "4,000 m

The grandest canyon in the solar system — 4,000 km long and up to 7 km deep.

11 Hebes Chasma

Lat 1.1° S · Lon 76.0° W · Elevation "5,000 m

An enclosed box canyon holding a massive layered interior mesa.

12 Ophir Chasma

Lat 4.0° S · Lon 72.0° W · Elevation "5,000 m

The northernmost chasma, with giant landslide deposits on its floor.

13 Candor Chasma

Lat 6.5° S · Lon 70.5° W · Elevation "5,500 m

Central canyon exposing layered sediments billions of years old.

14 Melas Chasma

Lat 11.5° S · Lon 70.0° W · Elevation "5,300 m

The deepest part of Valles Marineris, floored by landslides and sulfates.

15 Coprates Chasma

Lat 14.0° S · Lon 60.0° W · Elevation "5,000 m

An east-trending canyon segment lined with wall-layered deposits.

16 Ganges Chasma

Lat 8.0° S · Lon 48.0° W · Elevation "4,000 m

A wide chasma releasing catastrophic outflow channels eastward.

17 Capri Chasma

Lat 14.0° S · Lon 47.0° W · Elevation "3,500 m

A chaotic canyon region with evidence of ancient catastrophic floods.

18 Eos Chasma

Lat 16.0° S · Lon 41.0° W · Elevation "3,000 m

The eastern end of the canyon system feeds into the outflow channels.

19 Gusev Crater

Lat 14.5° S · Lon 175.4° E · Elevation "1,500 m

Spirit rover's home — an ancient crater once connected to a river via Ma'adim Vallis.

20 Apollinaris Patera

Lat 9.0° S · Lon 175.0° E · Elevation 5,000 m

A broad, old volcano north of Gusev with a large summit caldera.

21 Ma'adim Vallis

Lat 21.0° S · Lon 183.0° E · Elevation " 2,000 m

A 700 km outflow channel that likely spilled water into Gusev Crater.

22 Gale Crater

Lat 5.4° S · Lon 137.8° E · Elevation " 4,500 m

Curiosity rover's landing site; its floor holds the layered mound Aeolis Mons.

23 Aeolis Mons

Lat 5.1° S · Lon 137.7° E · Elevation 5,500 m

Mount Sharp — a 5 km sediment stack inside Gale recording Mars' climate history.

24 Holden Crater

Lat 26.4° S · Lon 325.0° E · Elevation " 1,500 m

An ancient lake crater with well-preserved delta and alluvial fan deposits.

25 Eberswalde Crater

Lat 24.0° S · Lon 326.0° E · Elevation " 800 m

Home to one of Mars' clearest fossil river deltas — a former habitable lake.

26 Nili Fossae

Lat 23.0° N · Lon 78.0° E · Elevation " 500 m

A graben trough rich in clay minerals marking long-ago water alteration.

27 Jezero Crater

Lat 18.4° N · Lon 77.7° E · Elevation " 2,400 m

Perseverance rover's landing site on a fossil river delta — prime biosignature target.

28 Nili Patera

Lat 23.5° N · Lon 80.0° E · Elevation 1,000 m

A caldera within Syrtis Major where active wind-driven sand migration was measured.

29 Isidis Planitia

Lat 14.0° N · Lon 88.0° E · Elevation " 3,400 m

A vast ancient impact basin between Utopia and Syrtis Major plains.

30 Lyot Crater

Lat 50.5° N · Lon 29.0° E · Elevation " 2,000 m

A large northern crater exposing subsurface hydrated minerals via its deep ejecta.

31 Utopia Planitia

Lat 46.7° N · Lon 117.5° E · Elevation " 3,300 m

Viking 2's landing plain — ice-cemented ground beneath a frozen lowland.

32 Vastitas Borealis

Lat 70.0° N · Lon 100.0° E · Elevation "4,500 m

The low-lying northern plains that ring the polar ice cap.

33 Acidalia Planitia

Lat 49.0° N · Lon 25.0° W · Elevation "4,000 m

A dark, dust-laden northern plain marked by enigmatic boulder fields.

34 Chryse Planitia

Lat 27.0° N · Lon 40.0° W · Elevation "2,700 m

The golden plains where Viking 1 first touched down on Mars.

35 Ares Vallis

Lat 22.0° N · Lon 35.0° W · Elevation "3,000 m

Martian Pathfinder's landing flood channel carved by ancient catastrophic flows.

36 Mawrth Vallis

Lat 22.3° N · Lon 19.0° W · Elevation "2,000 m

One of Mars' most clay-rich valleys, exposing colorful altered bedrock layers.

37 Tiu Valles

Lat 25.0° N · Lon 28.0° W · Elevation "2,500 m

An outflow channel system scoured by torrents from the highlands.

38 Simud Valles

Lat 20.0° N · Lon 25.0° W · Elevation "2,000 m

A teardrop-island-strewn channel from ancient megafloods.

39 Kasei Valles

Lat 30.0° N · Lon 60.0° W · Elevation "2,500 m

One of the largest outflow channels, over 1,500 km of flood-carved terrain.

40 Shalbatana Vallis

Lat 9.0° N · Lon 43.0° W · Elevation "1,500 m

A deep, sinuous channel that once hosted a standing paleolake.

41 Nanedi Valles

Lat 10.0° N · Lon 48.0° W · Elevation "1,200 m

A meandering valley with cliff walls suggesting long-lived water flow.

42 Athabasca Valles

Lat 9.0° N · Lon 149.0° E · Elevation "2,000 m

A young, plate-textured lava channel in Elysium, possibly water-carved.

43 Elysium Mons

Lat 25.0° N · Lon 147.0° E · Elevation 14,000 m

The northern volcanic province's main shield, wreathed in younger lava flows.

44 Hecates Tholus

Lat 33.0° N · Lon 150.0° E · Elevation 5,000 m

An older, deeply eroded volcano showing lava-channel lobes on its flanks.

45 Albor Tholus

Lat 19.0° N · Lon 150.0° E · Elevation 4,000 m

A small, low-relief shield completing the Elysium volcanic trio.

46 Orcus Patera

Lat 14.0° N · Lon 180.0° E · Elevation "1,000 m

An elongated, enigmatic depression between Elysium and Tartarus.

47 Hellas Planitia

Lat 42.0° S · Lon 70.0° E · Elevation "7,152 m

A colossal impact basin 2,300 km across and 7 km deep — one of Mars' deepest spots.

48 Argyre Planitia

Lat 49.0° S · Lon 43.0° E · Elevation "3,000 m

A ringed impact basin ringed by rugged mountains, once possibly filled with ice.

49 Newton Crater

Lat 41.0° S · Lon 159.0° E · Elevation "5,000 m

A 300 km impact basin whose walls show recurring dark slope streaks.

50 Lowell Crater

Lat 52.0° S · Lon 81.0° E · Elevation "3,500 m

A large southern crater near the Argyre rim, showcasing complex ejecta.